Deploying OpenNebula Cloud Platform on ESXi

Capstone Project

[HPC - 9123]

Submitted Date: 9.2.2024

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Document History

Date	Version	By	Remarks
19.1.2024	0.1	Khant Hmue Yeon Myat Eain May Thin Cho Phyo Zayar Kyaw Ye Htet Maung Maung	First version
25.1.2024	0.2	Group 2	Draft version
2.2.2024	0.3	Group 2	Draft version
6.2.2024	1.0	Group 2	Final Version

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Acknowledgement

This project would not have been possible without the guidance and support of many people, and we are deeply grateful for the continuous assistance we received throughout the entire journey. First and foremost, we express our heartfelt appreciation to our esteemed teachers, Dr. Aye Myat Myat Paing, Dr. Myat Pwint Phyu, Dr. Thandar Aung, Dr. Nwe Nwe Myint Thein from University of Information Technology who provided us with the opportunity to do this project on this topic. We also extend our sincere gratitude to our friends, who offered their help and expertise when needed. Their collaborative spirit and willingness to go the extra mile were crucial in ensuring we met our deadlines and produced our best work.

Project Summary

In summary, the successful implementation of the project aimed at creating a robust cloud infrastructure using OpenNebula has significantly elevated our organization's capabilities in managing and delivering IT services. Integrating the vCenter cluster into OpenNebula has proven instrumental in optimizing storage, compute, and network resources, thereby enhancing the overall efficiency and scalability of the cloud environment. The seamless coordination between OpenNebula and vCenter has provided a diverse range of services, ensuring that the organization remains agile in meeting evolving business demands. This project's success not only signifies a technological milestone but also underscores our commitment to staying at the forefront of innovative solutions for a dynamic and competitive IT landscape.

1. Introduction

Efficient resource utilization plays a critical role in businesses seeking optimal performance and cost effective and traditional infrastructures lack resource utilization. In the project, the IT infrastructure is deployed configuring two ESXi hosts, deploying vCenter Appliance and creating cluster pooling computing resources from two hosts into a single cluster. The OpenNebula platform is integrated for resource management and using its features.

1.1 Objective

The primary objective of this project is to establish a resilient and efficient resource utilization cloud infrastructure integrating with OpenNebula for comprehensive management of storage, compute, and network resources. The goal is to harness the combined capabilities of OpenNebula and vCenter to create a seamlessly integrated environment that would enable streamlined deployment and management of virtualized services. By aligning these technologies, the goal of the project is to enhance the overall agility, scalability, and performance, resource utilization of the cloud infrastructure, ultimately ensuring a robust foundation for delivering a variety of IT services.

1.2 Project Outcomes

In summary, the successful implementation of the project aimed at creating a robust cloud infrastructure using OpenNebula has significantly elevated our organization's capabilities in managing and delivering IT services. Integrating the vCenter cluster into OpenNebula has proven instrumental in optimizing storage, compute, and network resources, thereby enhancing the overall efficiency and scalability of the cloud environment. The seamless coordination between OpenNebula and vCenter has provided a diverse range of services, ensuring that the organization remains agile in meeting evolving business demands. This project's success not only signifies a technological milestone but also underscores our commitment to staying at the forefront of innovative solutions for a dynamic and competitive IT landscape.

2. What is OpenNebula?

OpenNebula is an open-source cloud computing platform that allows the management of virtualized data centers. OpenNebula combines virtualization and container technologies with multi-tenancy, automatic provision and elasticity to offer on-demand applications and services. OpenNebula supports various virtualization technologies, including KVM, LXD, and VMware. It provides users and administrators with a choice of interfaces. OpenNebula is known for its simplicity, flexibility, and power, making it a turnkey solution for constructing private clouds and managing data center virtualization. OpenNebula is widely used by industry and research leaders, attesting to its maturity and reliability with over a decade of software releases and numerous enterprise deployments.

2.1 What is vOneCloud?

vOneCloud is an open-source cloud management platform designed to simplify the deployment and management of virtualized infrastructure. One of its key advantages lies in its seamless integration with widely used open-source virtualization technologies, such as OpenNebula and KVM. This integration allows users to efficiently orchestrate, monitor, and automate their cloud environments, promoting flexibility and scalability. vOneCloud's user-friendly interface streamlines the management of virtual machines, enabling administrators to easily provision and control resources while optimizing performance. It also facilitates the importation of

resources, including virtual machines, networks, and storage, allowing seamless integration and utilization of existing VMware infrastructure. vOneCloud streamlines virtualized resource management, making it an accessible and efficient solution for VMware-centric environments with features such as template management, resource pools, and scalability options. Overall, vOneCloud stands out as a versatile and accessible solution for organizations seeking efficient and scalable cloud management.

2.2 Services provided by OpenNebula

Services provided by OpenNebula collectively enable users to build, deploy, and manage scalable and resilient cloud infrastructures while ensuring security, efficiency, and flexibility in managing virtualized workloads.

VM Management

OpenNebula enables users to easily create, deploy, manage, and monitor virtual machines (VMs) within their cloud infrastructure. Users can define VM templates with specific configurations (such as CPU, memory, disk, and network settings) and deploy VM instances based on these templates. OpenNebula offers features for lifecycle management of VMs, including start, stop, suspend, resume, and delete operations.

Storage Management

OpenNebula provides storage management capabilities for provisioning and managing storage resources within the cloud environment. Users can configure different types of storage, such as local disks, SAN, NAS, or object storage, and allocate storage resources to VM instances as needed. OpenNebula supports features like disk snapshots, cloning, and resizing to efficiently manage storage resources and meet the storage requirements of virtualized workloads.

Network Management

OpenNebula offers comprehensive network management features for configuring and managing network resources within the cloud infrastructure. Users can define virtual networks with specific network configurations, including VLANs, bridges, IP addressing, and network isolation. OpenNebula supports features like virtual routers, port forwarding, firewall rules, and load balancing to enable flexible and scalable network configurations.

High Availability

OpenNebula supports high availability (HA) configurations to ensure the continuous availability of virtualized workloads in the event of hardware failures or disruptions. HA features include VM migration, automatic recovery, and failover mechanisms that dynamically move VM instances to healthy hosts to minimize downtime and ensure service continuity.

Fault Tolerance

OpenNebula provides fault tolerance mechanisms to enhance the resilience and reliability of virtualized workloads against software failures or unexpected crashes. Fault tolerance features include automatic recovery, VM monitoring, and self-healing capabilities that detect and respond to VM failures, ensuring quick recovery and minimal disruption to services.

Scalability

OpenNebula offers scalability features to dynamically scale virtualized workloads based on changing demands and resource requirements. Users can vertically scale VM instances by adjusting CPU, memory, or disk resources, as well as horizontally scale by deploying multiple instances of the same VM template across the cloud infrastructure.

Self-Service Portal

OpenNebula provides a self-service portal that allows users to easily provision, manage, and monitor their virtualized workloads without requiring extensive technical expertise. The self-service portal offers an intuitive web-based interface for users to perform various tasks, including VM deployment, resource allocation, and monitoring of cloud resources.

API Integration

OpenNebula offers comprehensive APIs and SDKs for integrating with external systems, tools, and automation frameworks. APIs enable programmatic access to

OpenNebula's functionality, allowing users to automate cloud management tasks, integrate with third-party tools, and develop custom applications and workflows.

Monitoring and Reporting

OpenNebula provides monitoring and reporting capabilities to track the performance, utilization, and health of cloud resources. Users can monitor VM metrics, resource utilization, network traffic, and other performance indicators through built-in monitoring tools and integration with external monitoring solutions. OpenNebula also offers reporting features to generate customized reports on resource usage, billing, and compliance for better visibility and governance.

Security and Authentication

OpenNebula supports security features and authentication mechanisms to ensure the confidentiality, integrity, and availability of cloud resources. Users can implement role-based access control (RBAC), authentication methods (such as LDAP, Active Directory), and granular permissions to control access to cloud resources and enforce security policies. OpenNebula also offers features for encryption, secure communication (SSL/TLS), and compliance with security standards to protect sensitive data and ensure regulatory compliance.

2.3 OpenNebula Architecture

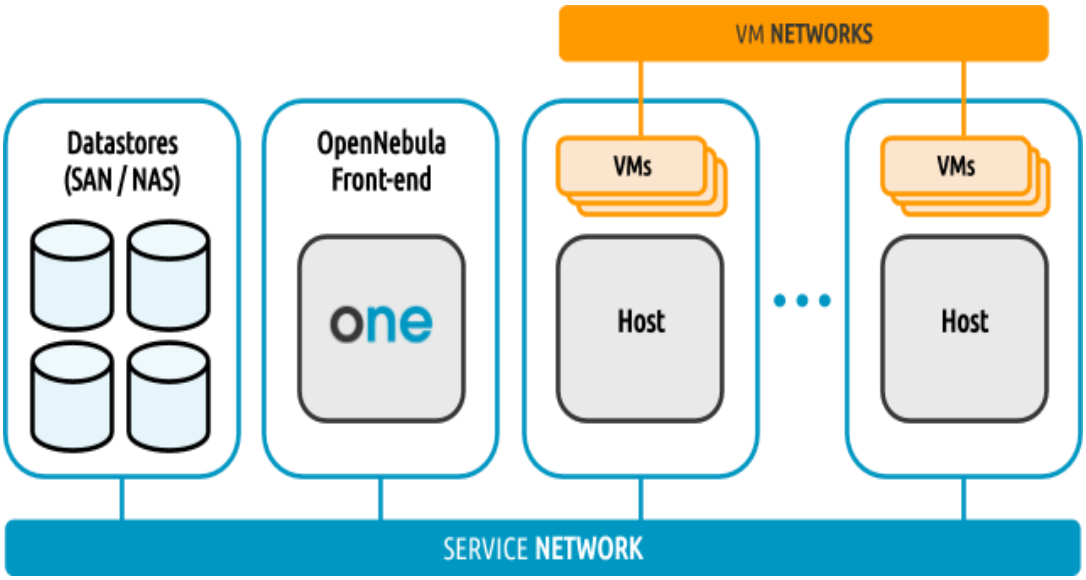

Figure 1 OpenNebula Architecture

OpenNebula is a freely available software that enables the creation of private, public, and hybrid clouds. It follows a traditional cluster-style structure consisting of a front-end, a datastore, a network, and a group of hypervisor-enabled nodes for executing and managing virtual machines (VMs). Figure-1 illustrates the key components, including the front-end, nodes, datastore, and the management daemon (ONED). The front-end runs the management daemon and cluster services, while the nodes function as hypervisor-enabled hosts capable of executing VMs. The datastore, which can be various storage systems like SAN and NAS servers, is utilized to store disk images for VMs. OpenNebula utilizes at least two distinct physical networks: a service network for front-end management and hypervisor-related tasks, and an instance network designed to provide network connectivity to VMs. OpenNebula supports multiple hypervisors, namely Xen,

KVM, and VMware ESX. Users can manage this cloud platform through either the Command Line Interface (CLI) or the SunStone web interface.

3. Requirement Specification

The minimum requirement specifications of hardware and software for setting up of infrastructure are organized as below.

3.1 Hardware Requirement

Two Cluster Nodes

1. Compute Node 1 (Dell Server)
    - RAM – 24 GB
    - HDD – 3.3 TB
    - Processor Core – 8 core
2. Compute Node 2
    - RAM – 16 GB
    - HDD – 500 GB
    - Processor Core – 4 core
3. Other Requirements
    - Dual-port NIC cards QTY: 2
    - Fast Ethernet Cable QTY: 4

3.2 Software Requirement

For software requirements specification, the minimum requirements are

- Ubuntu 18(Operating System)
- ESXi 6.5 (Hypervisor)
- vCenter Server Appliance 6.5
- vOneCloud 6.8.0

4. Project Architecture

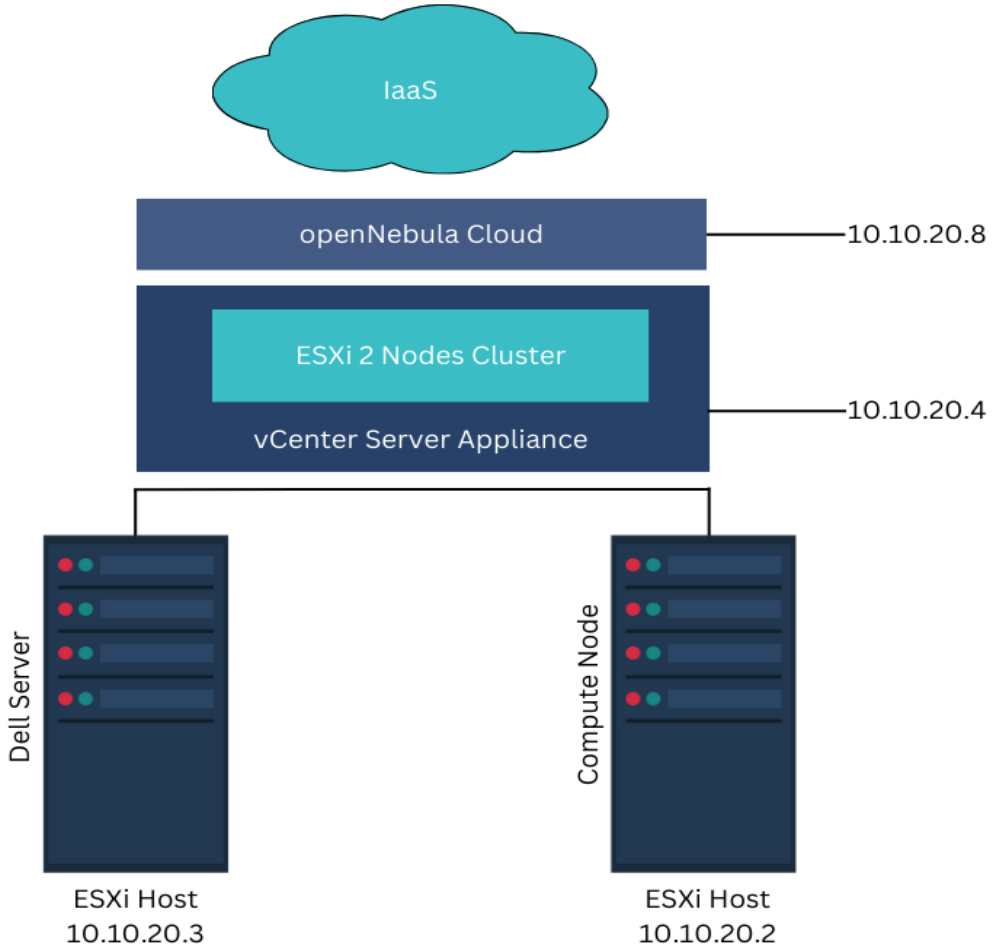

Figure 2 Project Architecture

The project provides a combination of OpenNebula and a vCenter cluster to create a powerful cloud setup is shown in Figure 2. This setup is great at handling storage, computing, and network tasks. OpenNebula acts as the coordinator, working closely with the vCenter cluster, which serves as a central hub for managing virtualized environments. Together, they make sure resources are used wisely, services can scale up smoothly, and everything runs efficiently. The architecture allows us to quickly set up and manage virtual services, making it easy for the organization to keep up with the latest tech needs.

5. Implementation of Project

The implementation of project is shown in this section.

5.1 Implementation of ESXi on two nodes

The two ESXi hosts which are used as compute and storage nodes are deployed.

Additional storage is added to one host which is Dell Server.

5.2 Implementation of VMware vCenter

First, installation of VMware vCenter from USB is deployed. One ESXi host is used to deploy vCenter Server with Embedded Platform Services Controller.

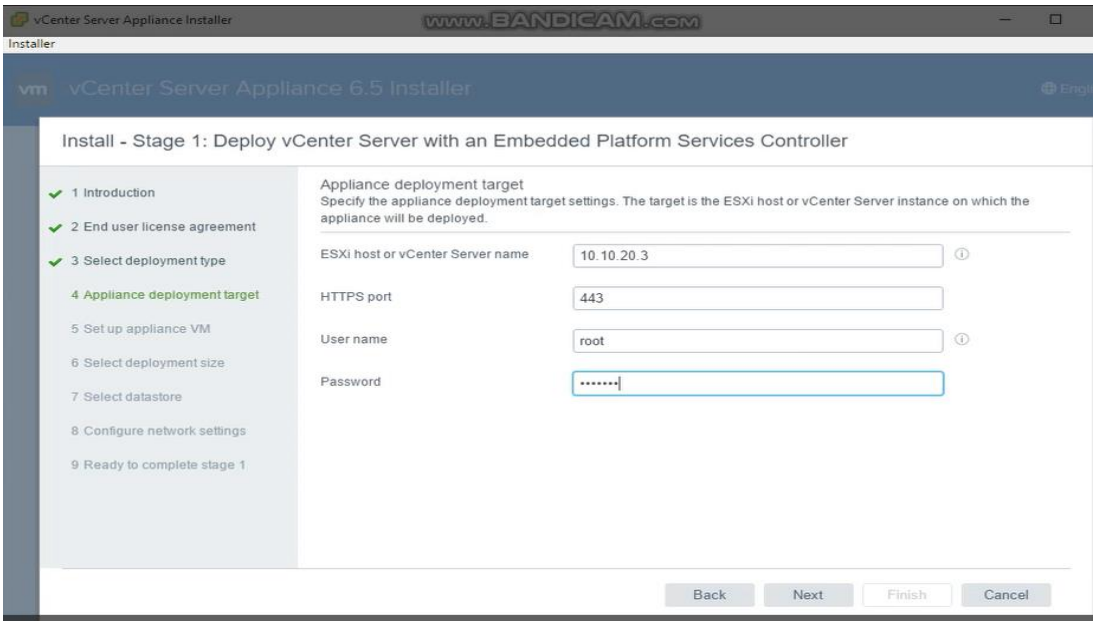

Figure 3 Verifying IP address of ESXi host

In Figure 3, Server name is defined by IP address of one ESXi host where vCenter Appliance is installed.

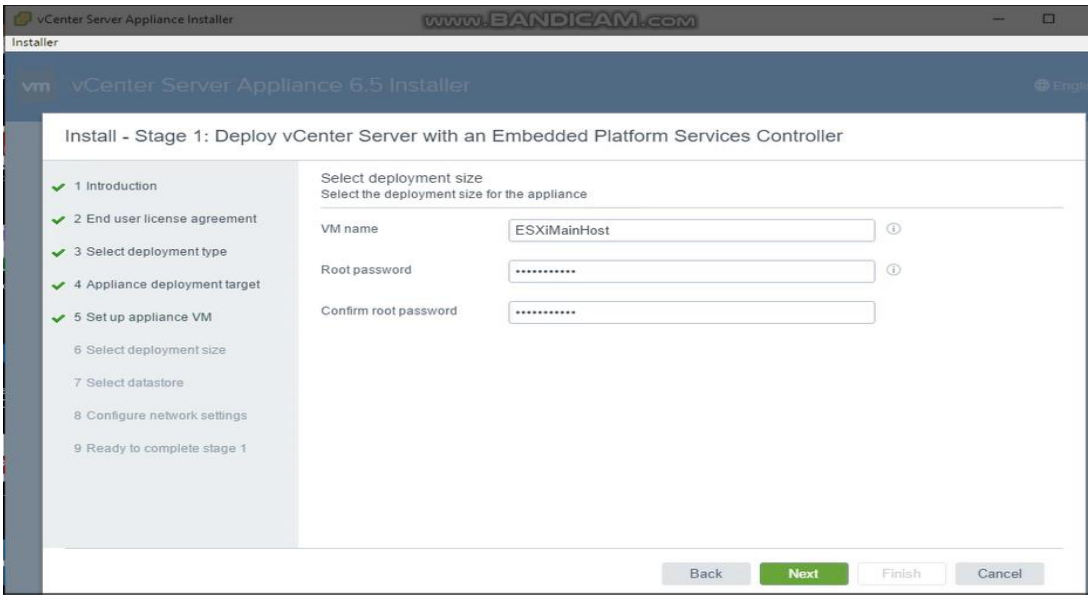

Figure 4 Setting password for vCenter

In Figure 4, the password of vCenter appliance is defined.

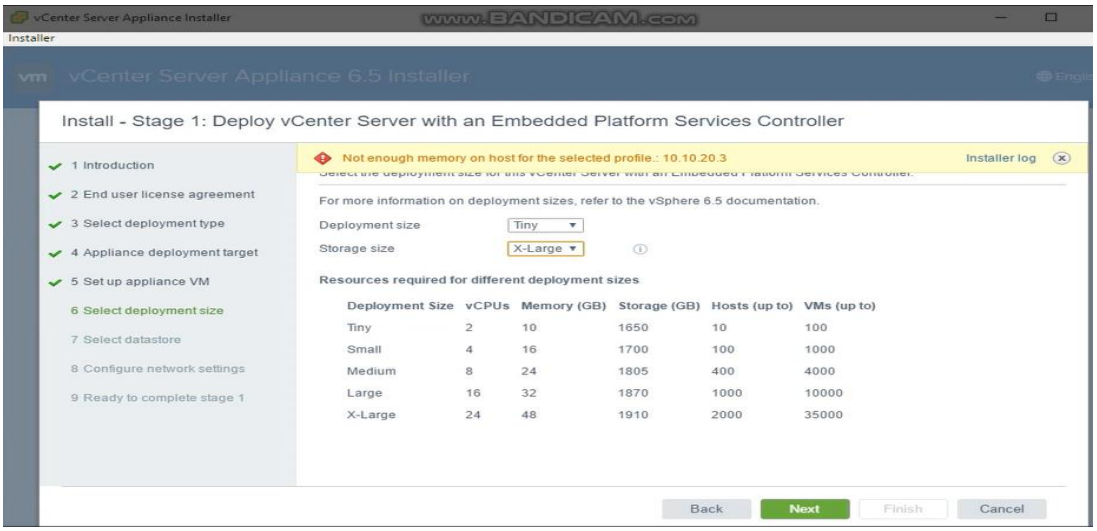

Figure 5 Choosing deployment size compatible hardware

Based on the available resources of the host, deployment size and storage size of vCenter is required to choose in Figure 5.

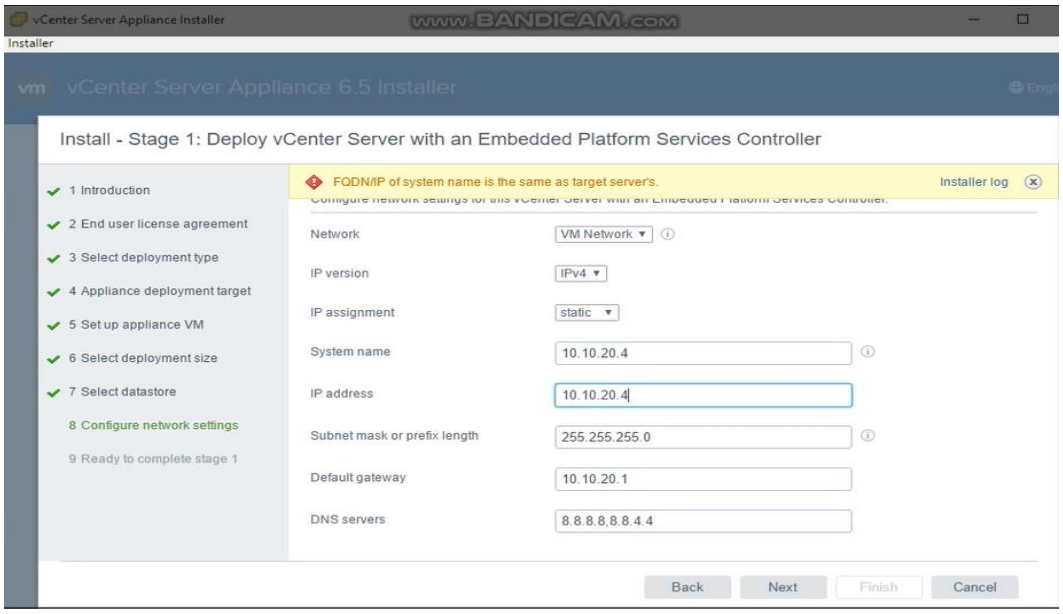

Figure 6 Setting IP address of vCenter

Figure 6 shows that IP address is defined for vCenter appliance that is included in defined network range. In the project, the network ranges from 10.10.20.2 to 10.10.20.9 is defined.

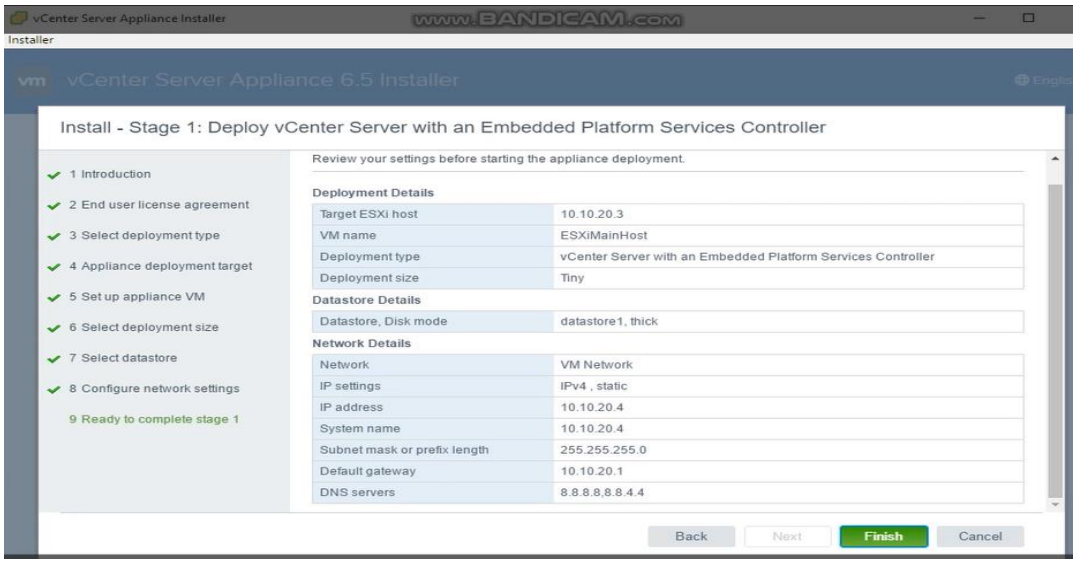

Figure 7 Summary of vCenter Appliance

Figure 7 illustrates the configuration setting of vCenter appliance deployment including which ESXi host is used for vCenter and deployment size.

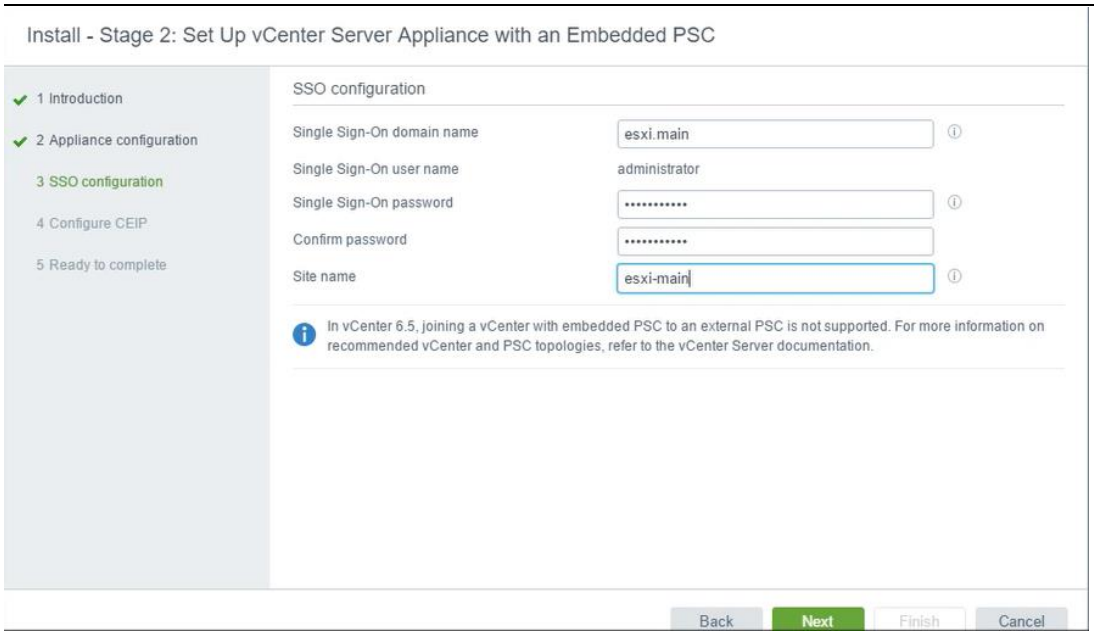

Figure 8 Setting up vCenter Server Appliance

The domain name needs to be assigned and site name shown in Figure 8.

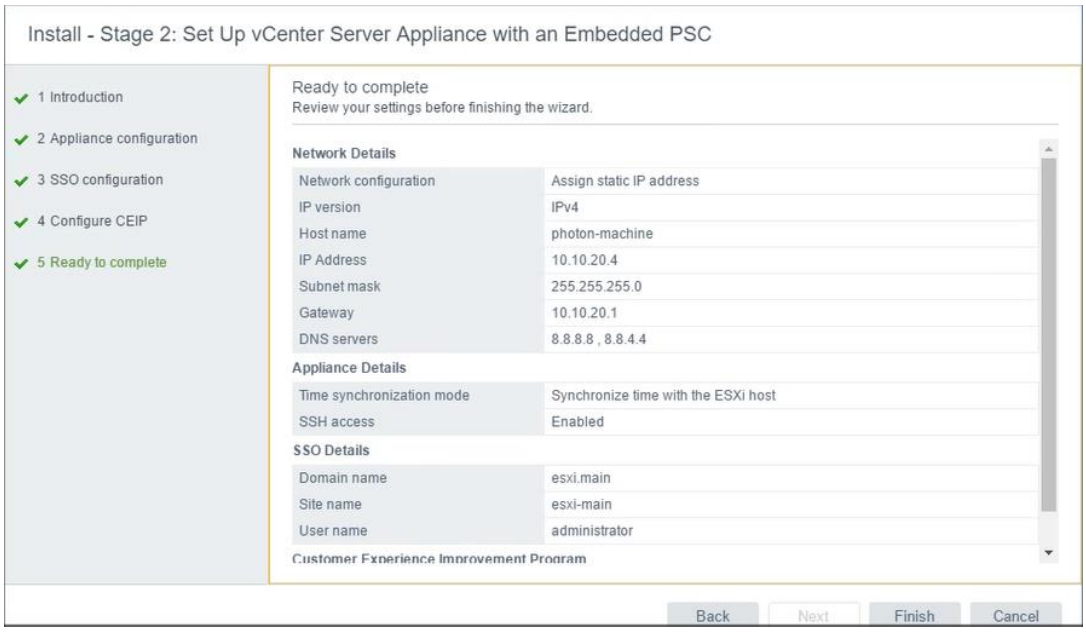

Figure 9 Complete Setting up of vCenter Appliance

Figure 9 demonstrates the complete set up for vCenter appliance including its IP address.

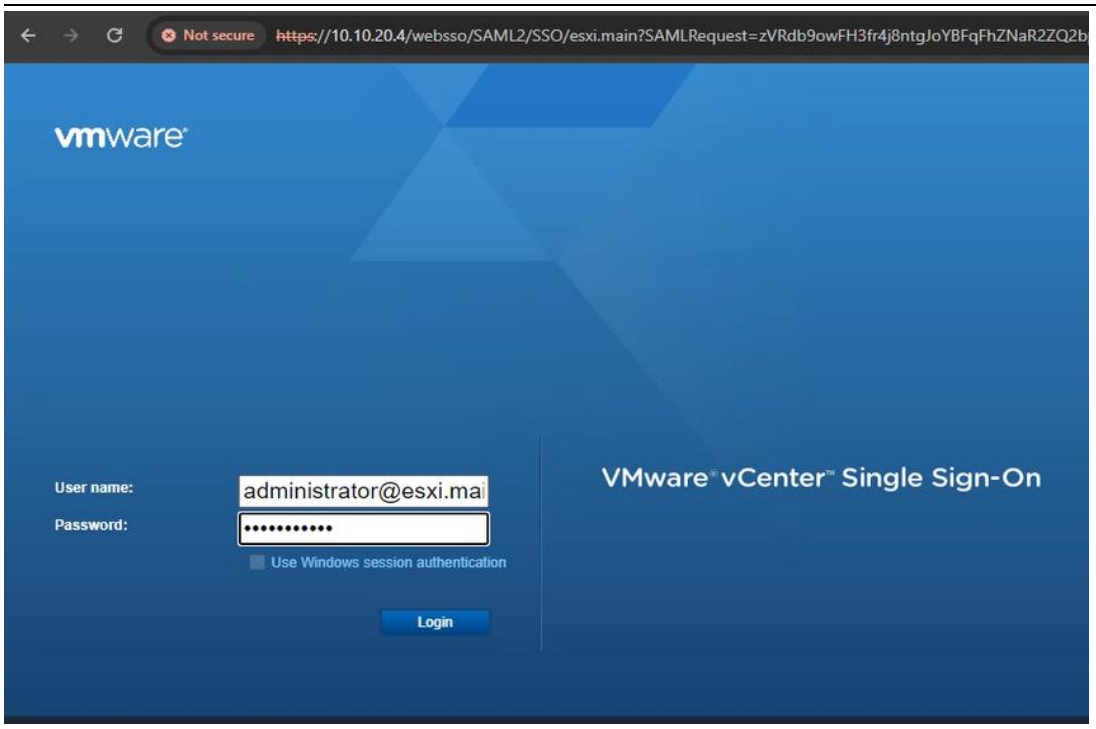

Figure 10 Interface of vCenter Appliance

The interface of vCenter is reached from the IP address of vCenter 10.10.20.4 for central management of the compute nodes after completing setting up of vCenter in Figure 10.

5.3 Datacenter Creation

A virtual data center is a container for all the inventory objects required to complete a fully functional environment for operating virtual machines and referred to as physical environment where resources are allocated.

In vCenter, Datacenter is created after right-click the vCenter Server object and select New Datacenter. By adding two ESXi hosts to Datacenter for complete environment for operating virtual machines. After right-click Datacenter and choose Add Host and fill IP address of one ESXi host.

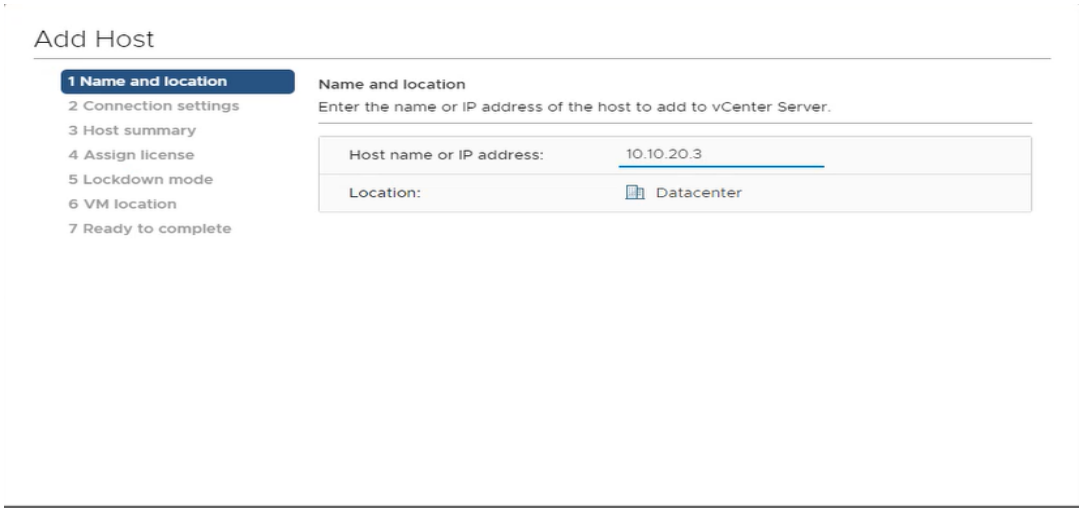

Figure 11 Adding host for Datacenter

IP address of ESXi host is defined in Host name section that includes in creation of Datacenter in Figure 11.

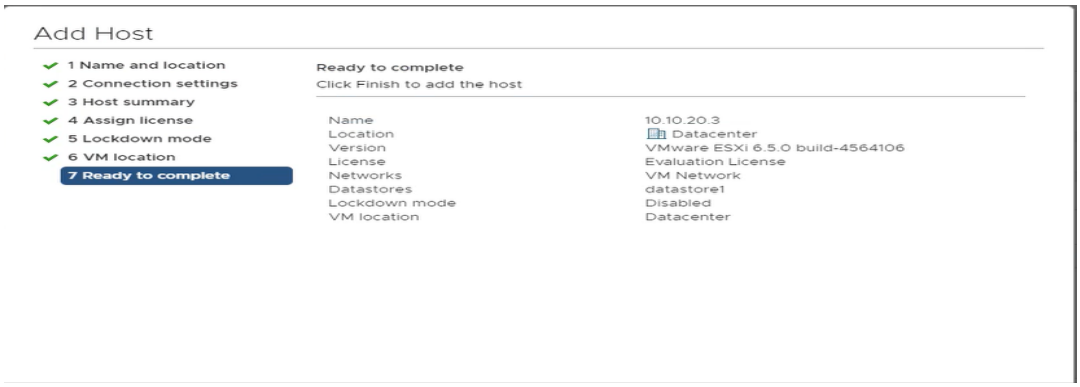

Figure 12 Summary of Datacenter

The summary of the host is shown in Figure 12.

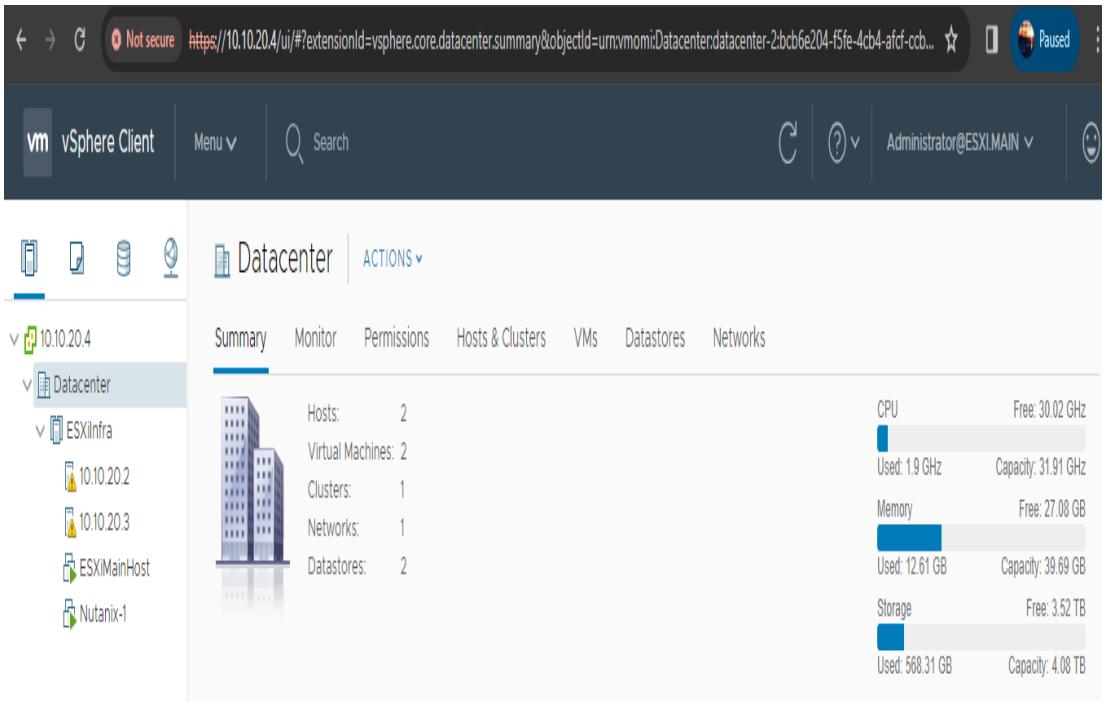

Figure 13 Datacenter Status

Figure 13 demonstrates the state of Datacenter including memory, CPU and storage.

5.4 Cluster Creation

Next, vCenter ESXi host cluster will be created. A cluster is a group of ESXi hosts where host resources are added to cluster becoming a resource pool. To create cluster, right-click on Datacenter and navigate New Cluster and create cluster.

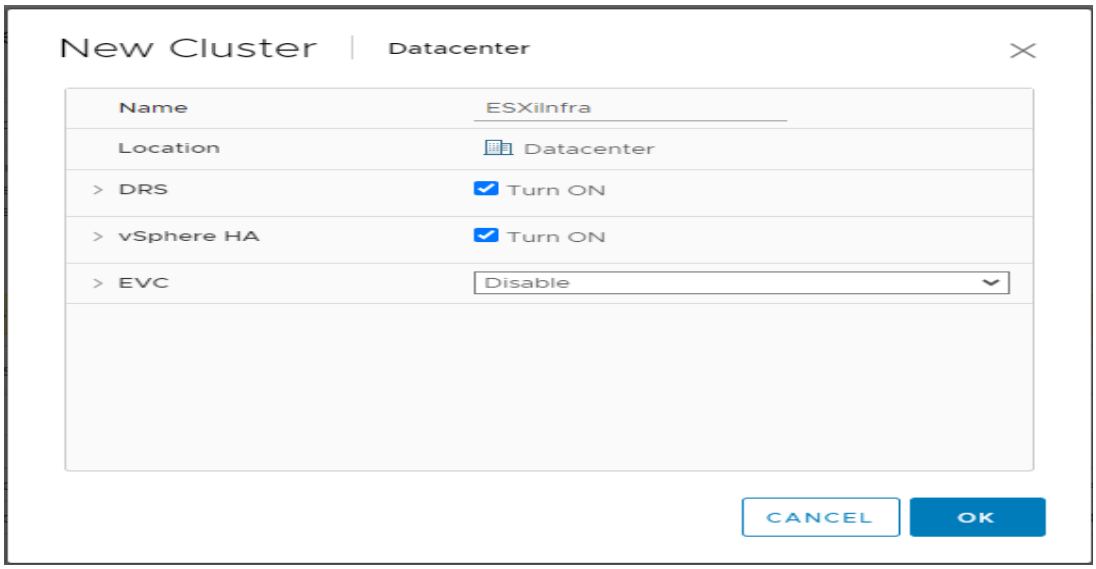

Figure 14 Cluster Creation Progress

Figure 14 presents creation of cluster "ESXiInfra", by adding the two ESXi hosts to cluster for pooling of physical resources.

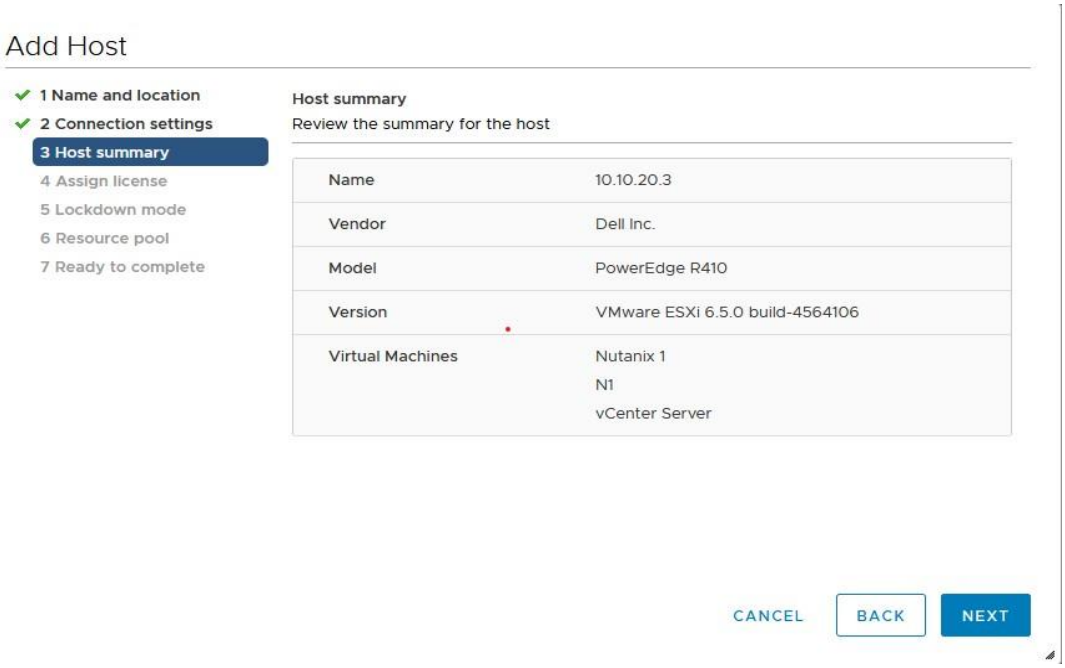

Figure 15 Host Summary in Cluster

After creating cluster "ESXiInfra", the two ESXi hosts are added to cluster. Figure 15 expresses the summary of one ESXi host.

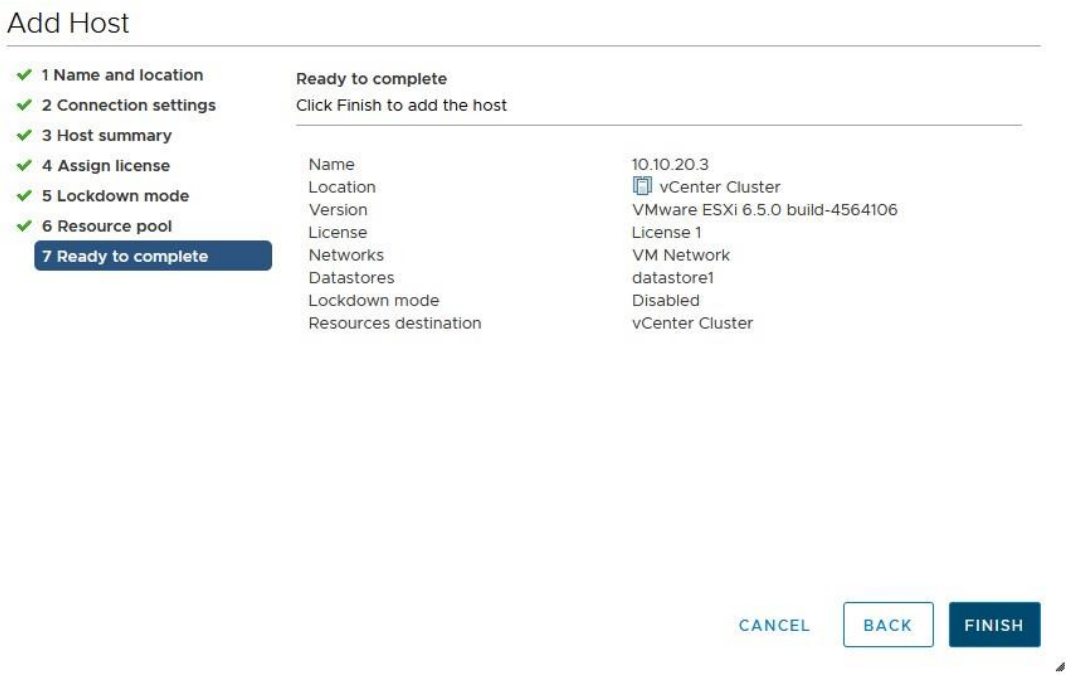

Figure 16 Host Summary in Cluster

The second host is added to cluster. The summary of second ESXi host is shown in Figure 16.

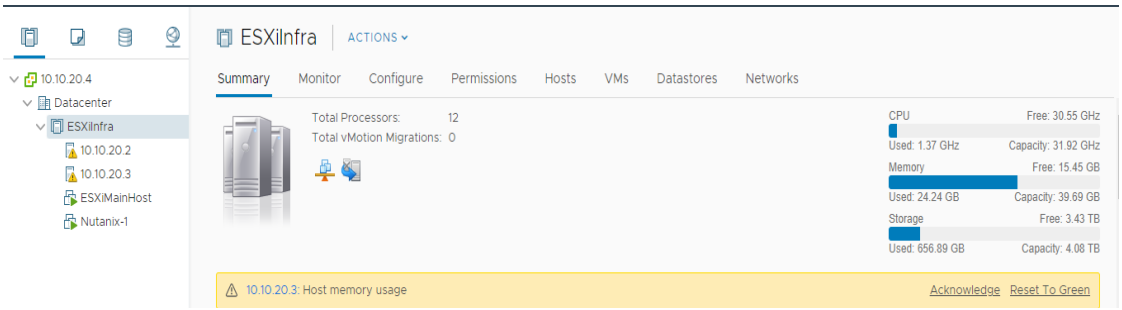

Figure 17 Cluster Status

The summary of the cluster, “ESXiInfra” including the state of using memory, CPU and storage of resources is presented in Figure 17.

5.5 OVF Template Creation in vCenter

The deployment of OpenNebula from vCenter is done.

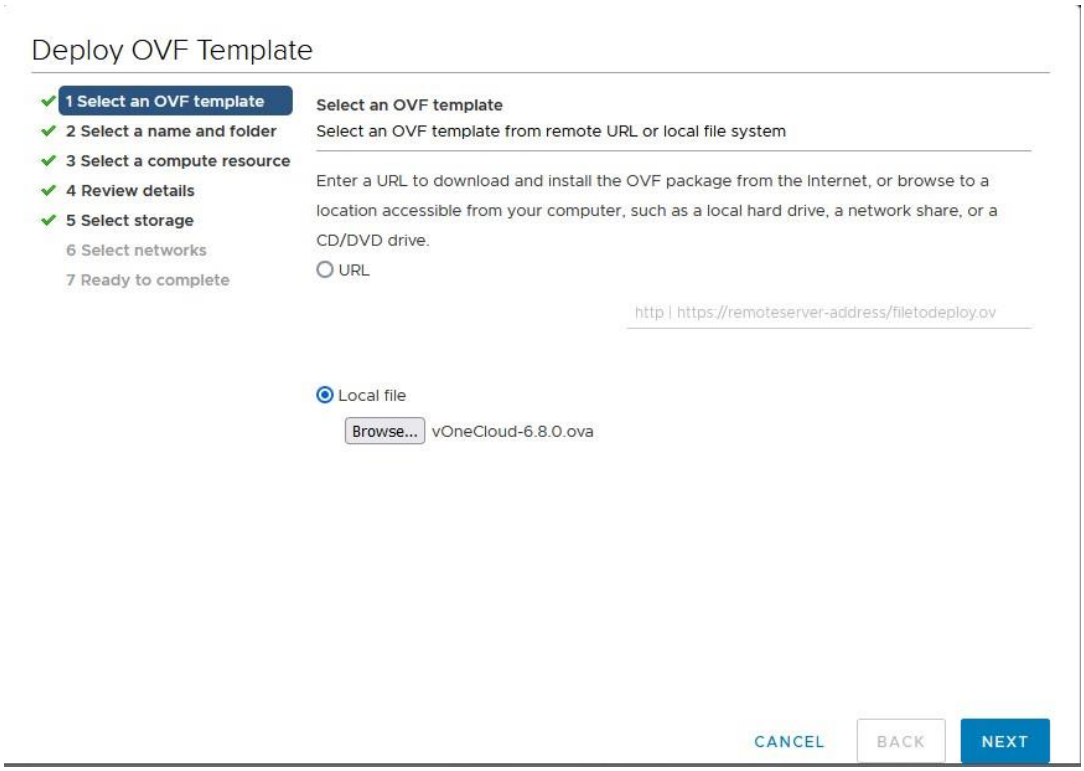

Figure 18 Selecting OVF Template

First, right-click on host or cluster and select "Deploy OVF template". The OVF Template that is required for installation of OpenNebula is represented on the above, Figure 18.

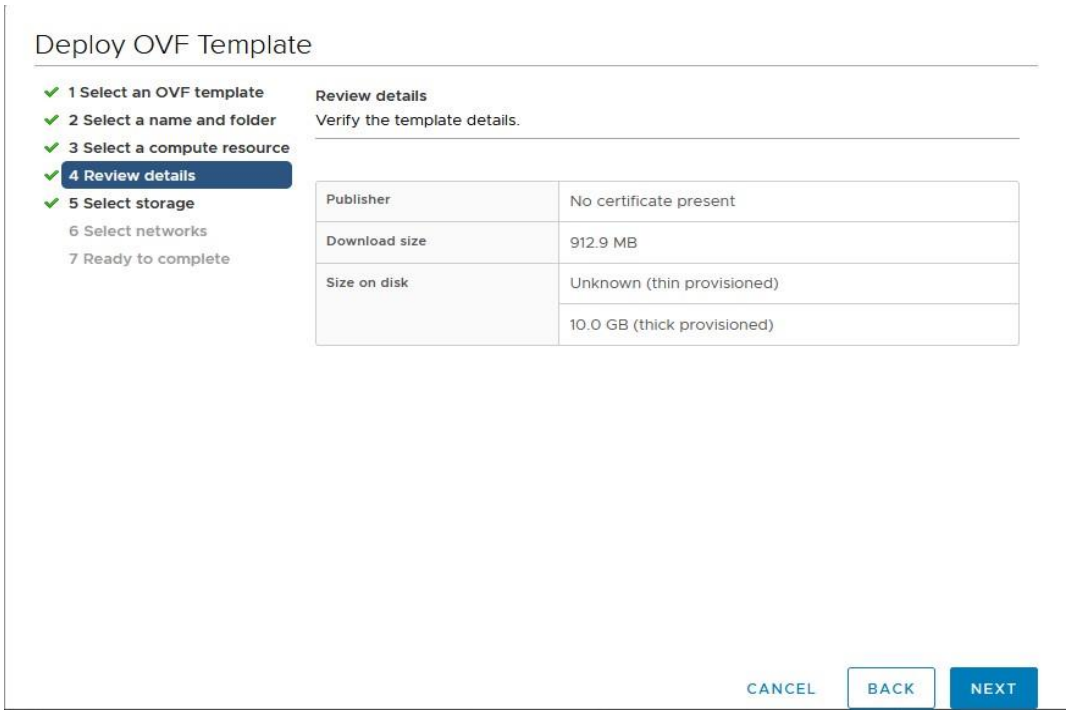

Figure 19 Template detail

Figure 19 is the detail of OVF Template.

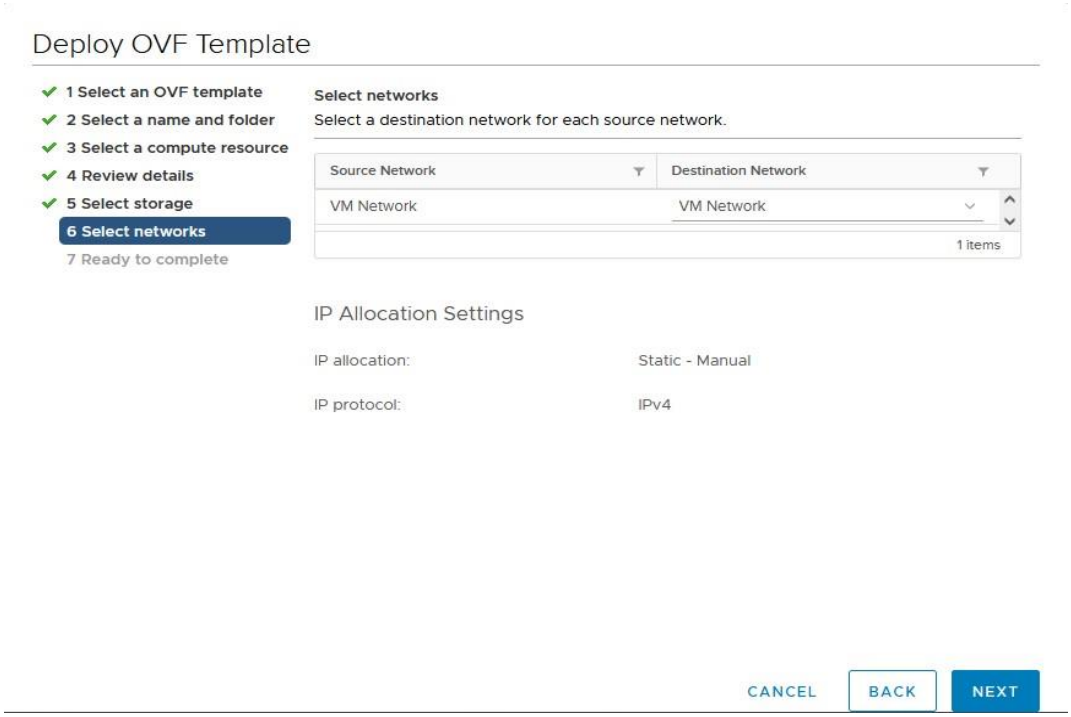

Figure 20 Network Selecting

Source Network is needed to choose VM Network demonstrated in Figure 20.

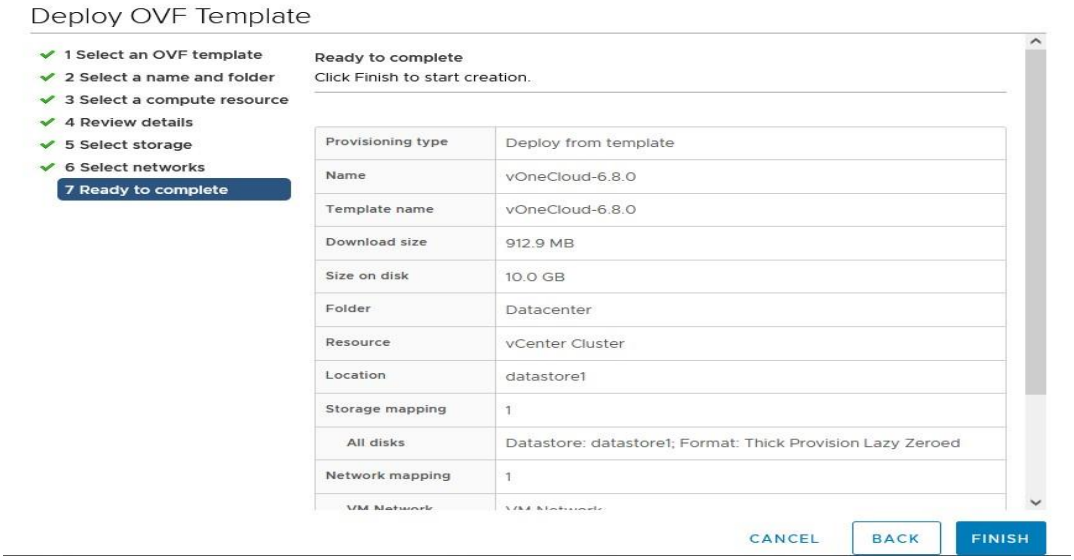

Figure 21 Summary of OVF Template

Figure 21 is the summary of OVF template.

5.6 Configuring for OpenNebula

After setting up OVF Template, navigate “vOneCloud-6.8.0” and power on.

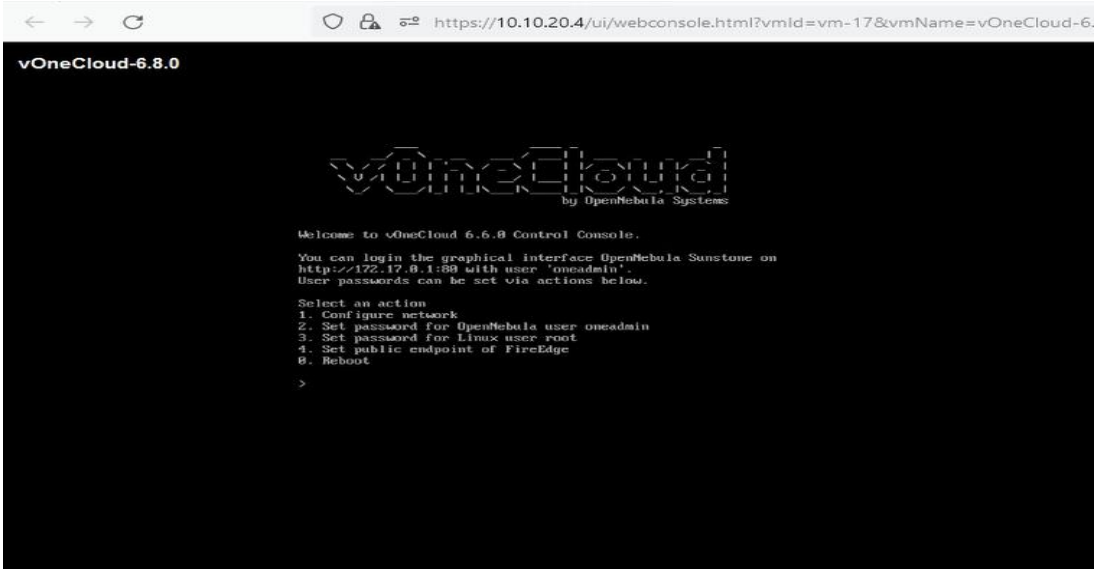

Figure 22 Booting vOneCloud

After booting vOneCloud, the configuration setting of OpenNebula is shown in Figure 22.

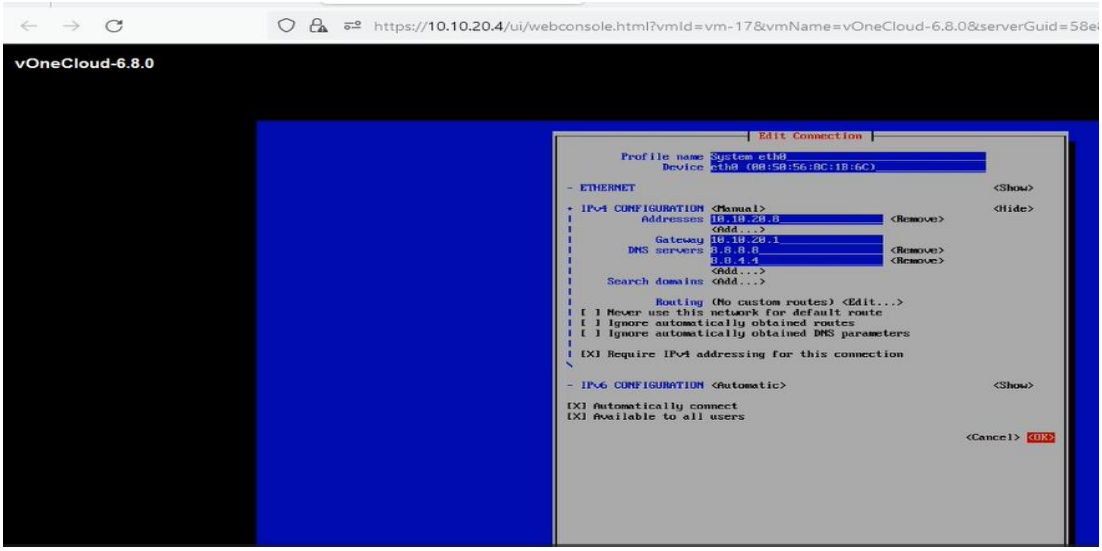

Figure 23 Configuring Network Connection

The Figure 23 illustrates the network configuration setting for OpenNebula and it is needed to be defined the IP address for accessing OpenNebula.

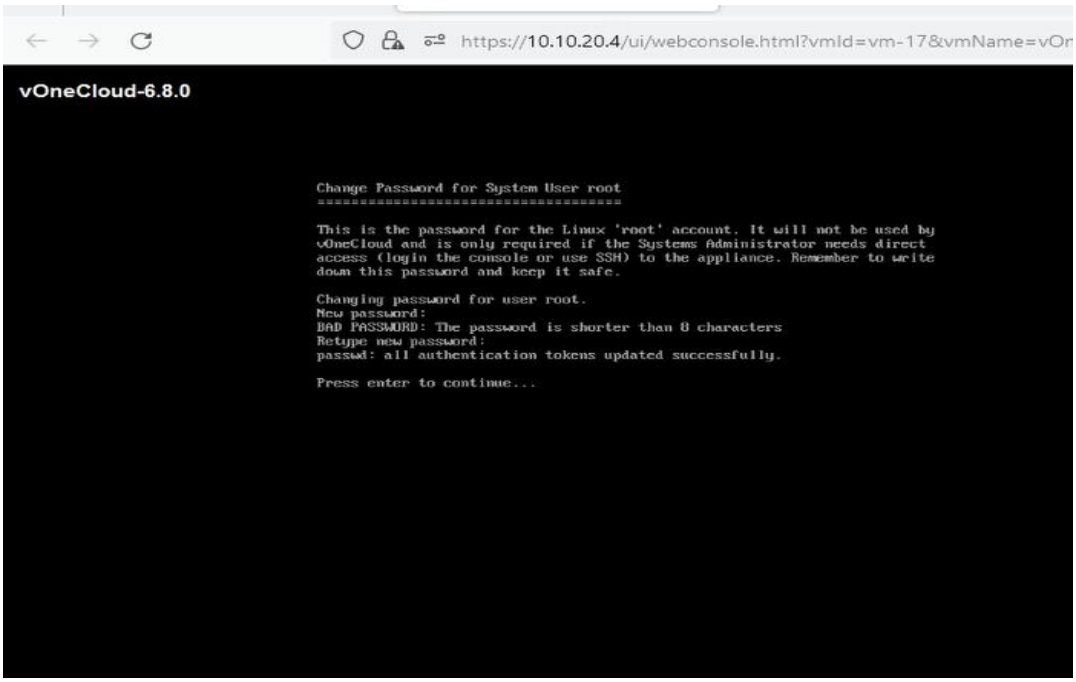

Figure 24 Verifying Password for OpenNebula

Figure 24 is the password of OpenNebula being verified.

Figure 25 Verifying Endpoint Address

Figure 25 verifies endpoint address. The endpoint address is needed to set IP address for accessing end user the interface of OpenNebula so IP address 10.10.20.8 that is existed in network range.

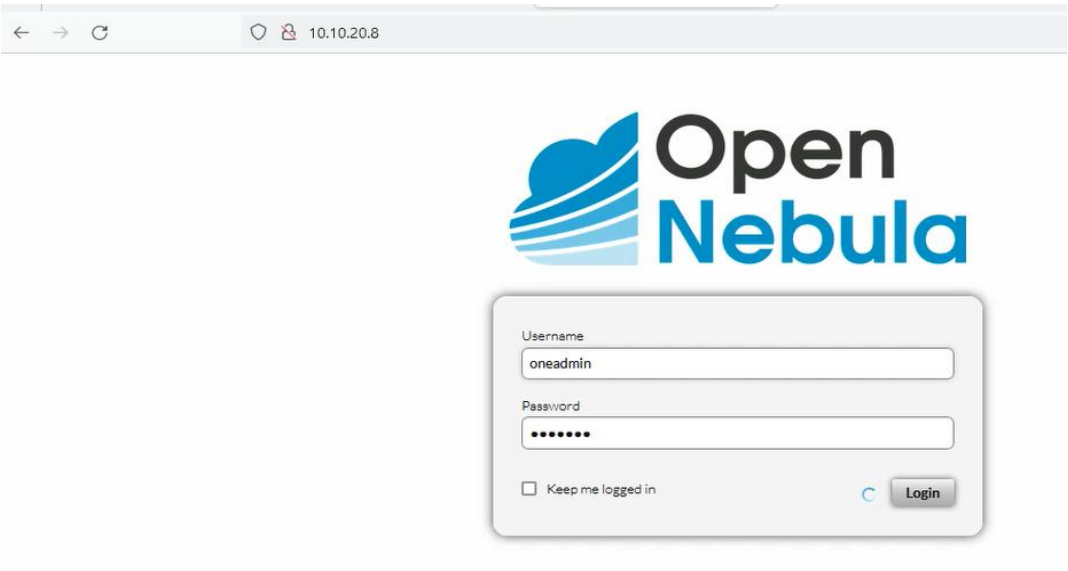

Figure 26 OpenNebula Interface

In Figure 26, after verifying required packages, the GUI of OpenNebula is ready to browse.

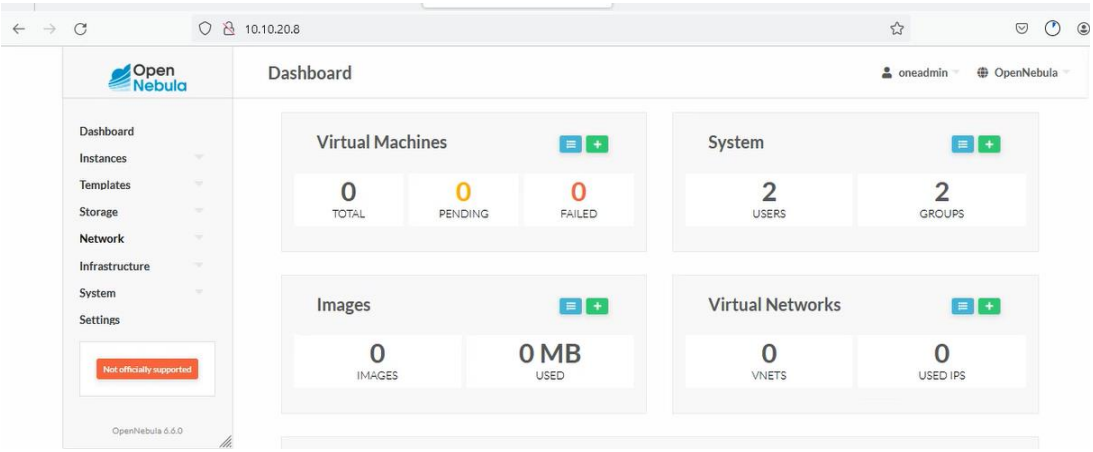

Figure 27 OpenNebula Dashboard

Figure 27 is the dashboard of OpenNebula.

6. Deploying IaaS on OpenNebula

This section illustrates the deployment of the project.

6.1 Cluster Deployment

To add the cluster that is created in vCenter, navigate “Host” and add the cluster from vCenter.

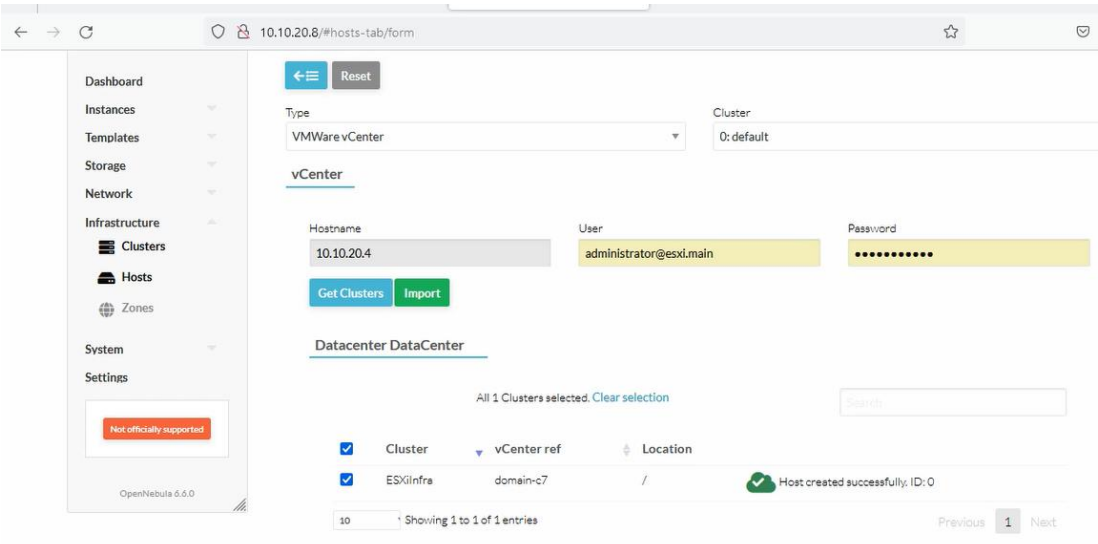

Figure 28 Adding Cluster from vCenter

Host name is needed to be the IP address of vCenter appliance because resources from vCenter is extended to OpenNebula for offering services which can be seen in Figure 28.

6.2 Network Configuration

After cluster creation, network configuration is needed to proceed. To make network configuration, navigate “Virtual Network” and select network of existing “ESXiInfra”.

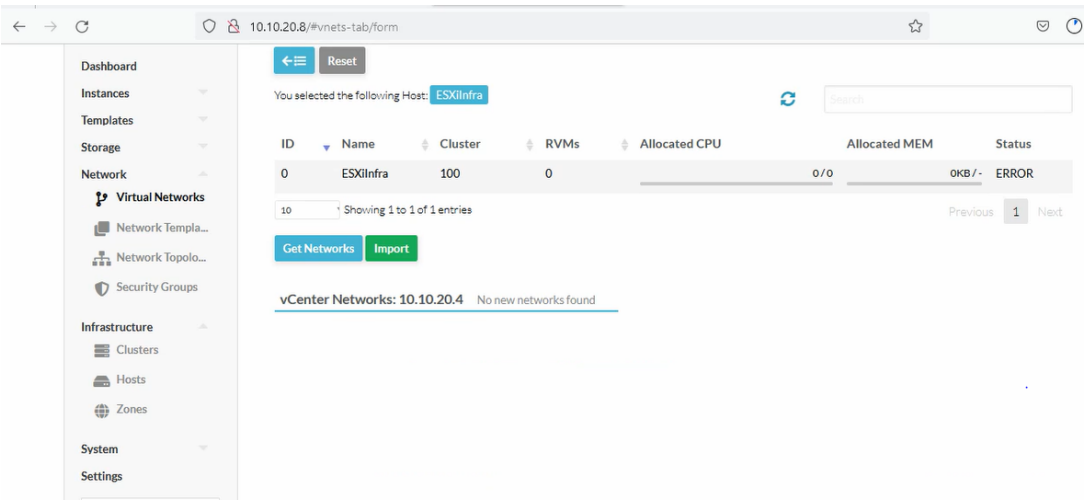

Figure 29 Network Configuration for Cluster

The Virtual Network is defined as the same network of vCenter Networks which is portrayed above, Figure 29.

6.3 Adding Storage to Cluster

The storage capacity from vCenter is added to cluster in OpenNebula and navigate "Datastore" and adding storage to cluster.

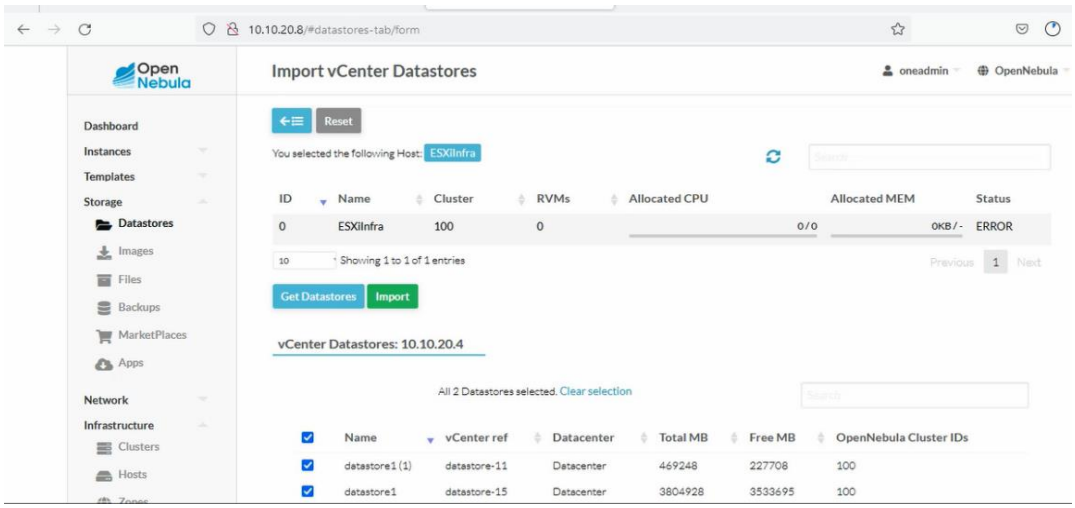

Figure 30 Storage Capacity of Cluster

Figure 30 illustrates the expansion of Cluster storage, extending from the vCenter datastore to provide scalable services.

6.4 Creating Virtual Machine

In OpenNebula Dashboard, click the "VMs" under the "Instances" Tab which appears in the Left side menu. The virtual machine is created by using images from OpenNebula Marketplace.

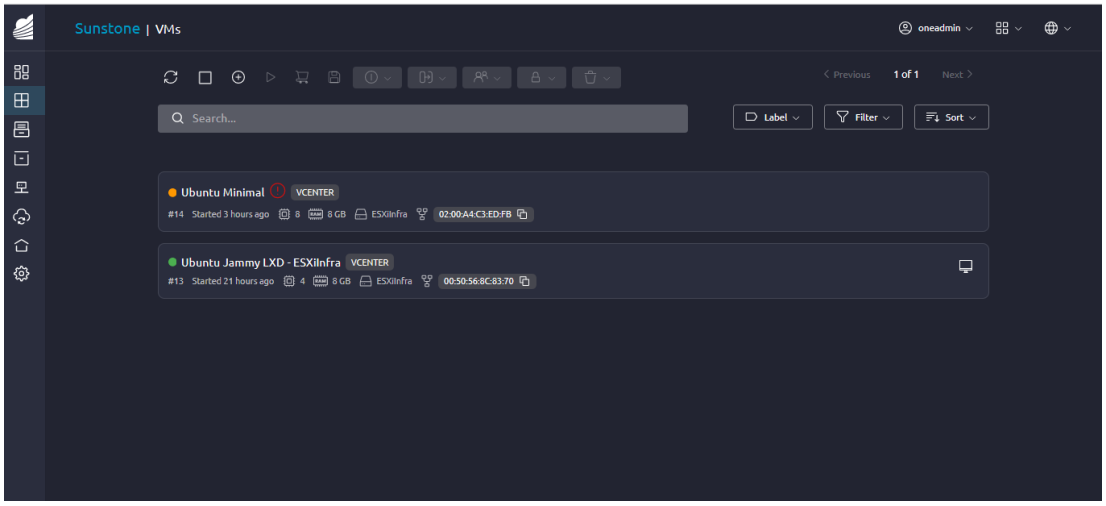

Figure 31 Virtual Machine Creation

In Figure 31, the virtual machine is shown that is created by using Ubuntu ISO file.

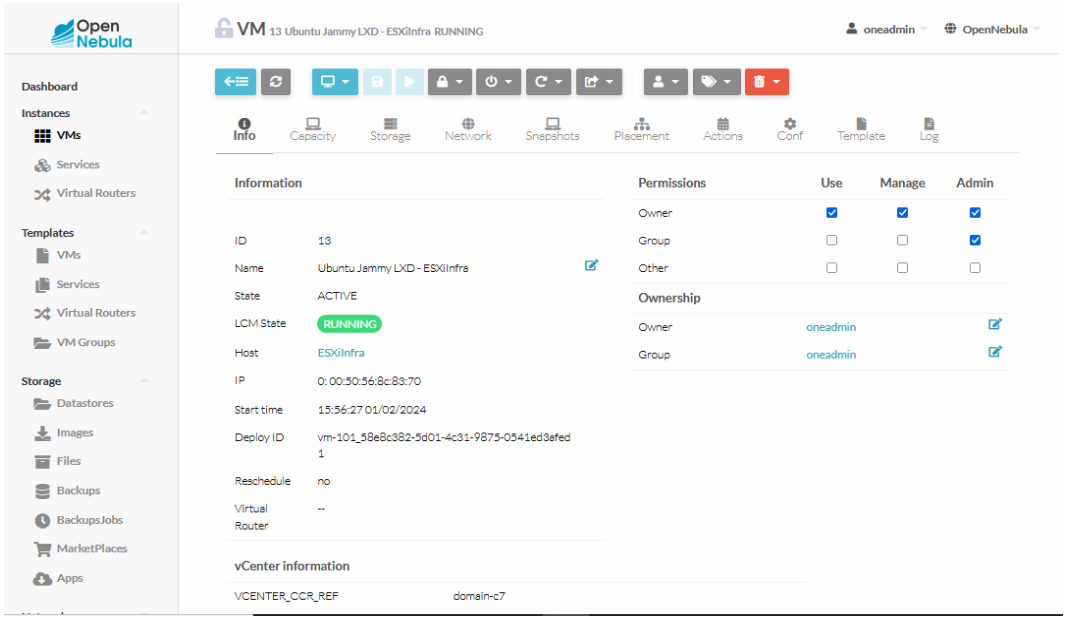

Figure 32 Virtual Machine Information

The Figure 32 shows the status of VM and its information.

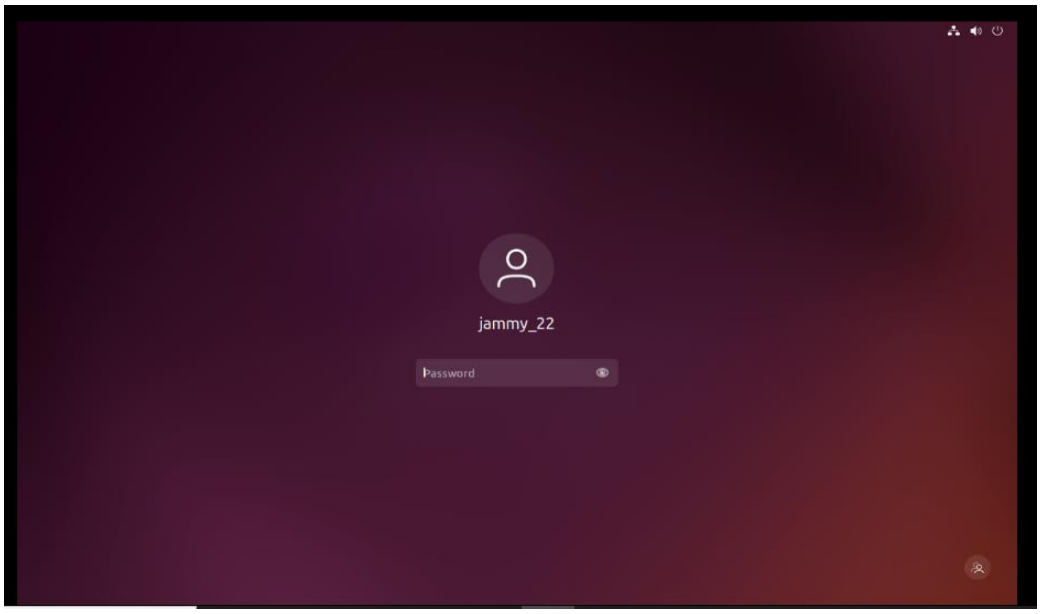

Figure 33 Virtual Machine

Figure 33 shows the virtual machine that is running in OpenNebula.

Figure 34 Network Condition

Figure 34 demonstrates the condition of network in virtual machine.

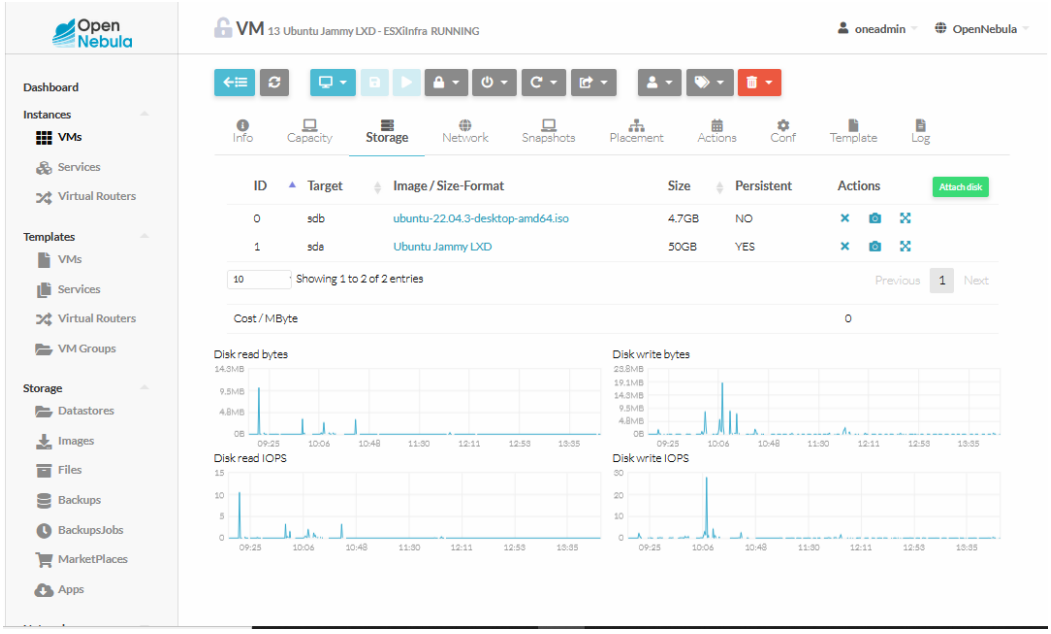

Figure 35 Storage

The storage of virtual machine is presented in Figure 35.

Figure 36 Capacity

The usage of capacity of virtual machine can be seen in Figure 36.

6.5 Providing Services

The services that are created in OpenNebula are shown.

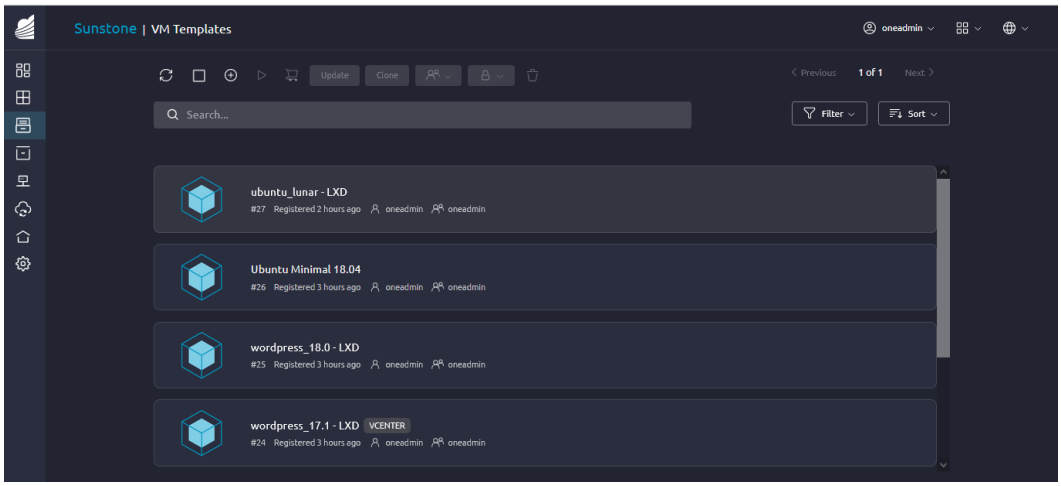

Figure 37 VM Templates

Figure 37 illustrates VM template that is used to instantiate a new virtual machine to perform the execution.

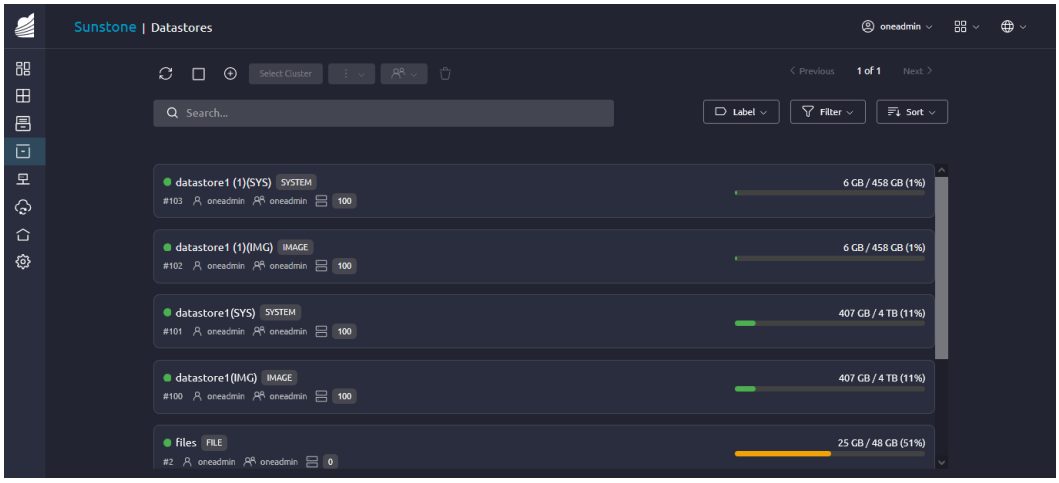

Figure 38 Datastores

Figure 38 explains storage capacity that is created in OpenNebula.

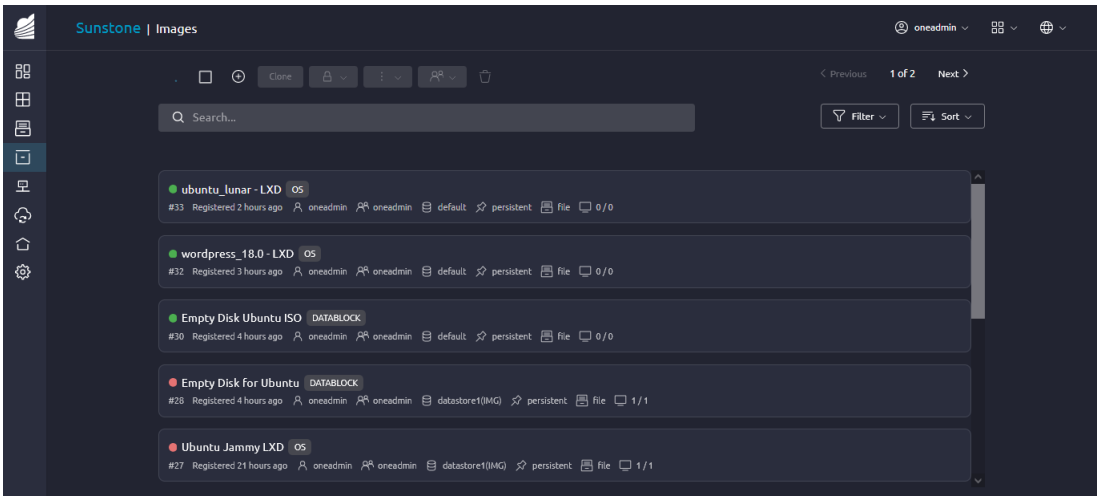

Figure 39 Images

Images for creation of virtual machines is demonstrated in Figure 39.

Figure 40 Virtual Networks

In Figure 40, the virtual networks that are created in OpenNebula.

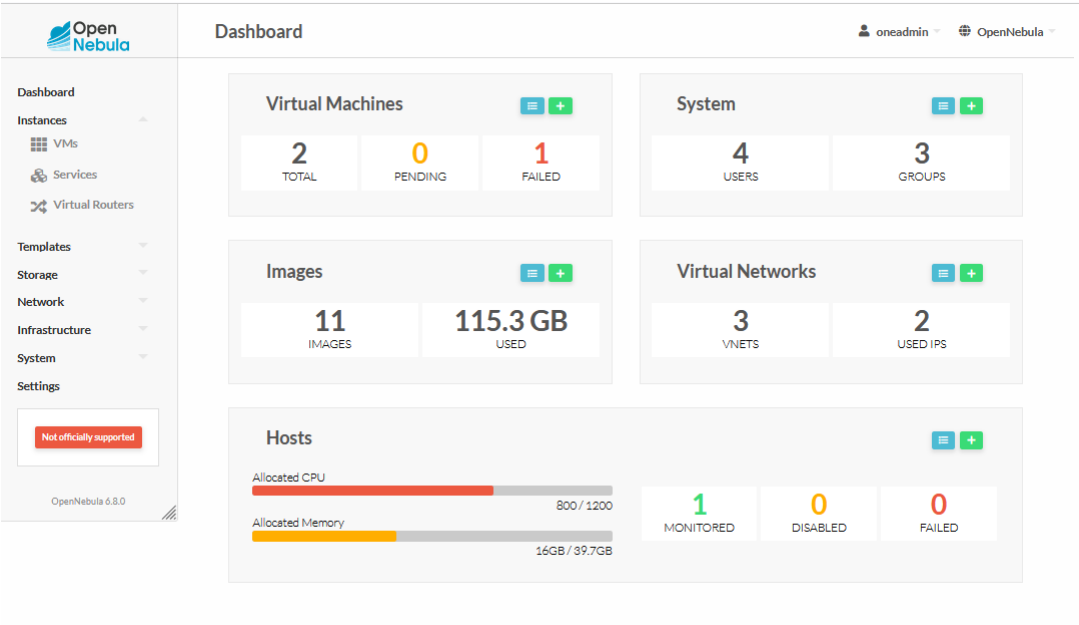

Figure 41 Dashboard

Figure 41 delivers the services that are created for providing IaaS to users.

6.6 Troubleshooting

During the implementation and deployment of infrastructure, the following errors are found.

6.6.1 Nutanix ISO File Error

In the installation of Nutanix, the error is “Failed to load phoenix ISO file”. This is error is because of the lack of SSD in hardware.

6.6.2 Nutanix Partition Error

The deployment of Nutanix Infrastructure is continued on Virtual Machine that is created in Cluster in VMware vCenter where virtual SSD can be created. The error is "Could not find Nutanix partition". This error is solved by selecting the SCSI controller while deploying Virtual Machine for Nutanix Installation.

6.6.3 Nutanix Controller VM Error

The Nutanix Controller VM plays crucial role in Nutanix Cluster for managing and controlling of components of Nutanix. In running of CVM, the error is "Fail to start domain NTNX CVM: Unexpectedly closed the monitor". This is error is because of insufficient of RAM in hardware. The installation of Nutanix file is made on Virtual Machine in vCenter so the running of vCenter Appliance takes 16 GB of RAM and the total of RAM is 32 GB. The running of Nutanix CVM requires 24 GB at least. Because of the lack of SSD in hardware and insufficient of RAM for running CVM, the deployment of cluster is changed to OpenNebula.

6.6.4 Network Card Disconnection

The standalone ESXi host (10.10.20.2) is disconnected and the network card of this host is disconnected. So, entering the configuration of network management setting and reconfigure network adapter.

6.6.5 Service Unavailable of vCenter Appliance

The interface of vCenter Appliance is not reachable because of error “503 Service Unavailable: Fail to connect to endpoint”. So, the time zone of two ESXi hosts is checked and updated with NTP server and time is synchronized with two ESXi hosts.

7. Conclusion

In the deployment of project, two ESXi hosts are deployed and combing these two nodes into a single cluster by deploying vCenter Appliance for resource utilization.OpenNebular cloud platform is integrated with ESXi two nodes cluster for creating IaaS services.The architecture of project illustrates the efficient resource utilization offering resource pooling of underlying hardware and integration with OpenNebula platform offering simple and flexible open-source solution to build private cloud.

7.1 Future Works

In the future endeavors, expanding the OpenNebula infrastructure will involve the seamless integration of new KVM and Linux nodes, enhancing the computational capabilities of our cloud environment. Further diversifying the cloud offerings, the aim is to include additional providers like AWS, Digital Ocean, and Azure, broadening the spectrum of services available to our users.

A significant aspect of the future work will also involve the implementation of container orchestration, allowing for efficient management and scaling of containerized applications. By incorporating tools like Kubernetes to ensure a more dynamic and responsive environment for deploying and running containerized workloads.

To strengthen the reliability and performance of the cloud, the introduction of a comprehensive monitoring and alert system is essential. Leveraging technologies

such as Prometheus and Grafana will enable real-time tracking of system metrics, proactive issue detection, and efficient incident response. This addition will contribute to the overall stability and resilience of the cloud infrastructure, ensuring a smooth user experience.

The future trajectory of the project encompasses the expansion of computational resources, integration with major cloud providers, incorporation of container orchestration, and the implementation of a robust monitoring and alert system. These initiatives collectively aim to fortify the cloud infrastructure, providing an agile and feature-rich platform for meeting the evolving needs of our organization.

7.2 References

[1]https://docs.vmware.com/en/VMware-vSphere/7.0/com.vmware.esxi.install.doc/GUID-93D0227B-E5ED-40B0-B8E2-71141A32EB00.html

[2]https://docs.vmware.com/en/VMware-vSphere/7.0/com.vmware.vcenter.install.doc/GUID-F06BA415-66D8-42CD-9151-701BBBCE8D65.html

[3]https://docs.vmware.com/en/VMware-vSphere/7.0/com.vmware.vsphere.vcenterhost.doc/GUID-6A30F824-3702-4580-94A2-C20CA8501547.html

[4] https://opennebula.io/discover/#opennebula_overview

[5]https://docs.opennebula.io/6.6/quick_start/deployment_basics/index.html

[6]https://docs.opennebula.io/6.6/quick_start/deployment_basics/try_opennebula_hosted.html

[7] https://docs.opennebula.io/6.6/installation_and_configuration/index.html

[8] https://docs.opennebula.io/6.6/vmware_cluster_deployment/index.html